

Multiple Choice (10 marks)

1. Calculate the Larmor frequency for a proton in a magnetic field of 1 T.
 - (a) 23.56 GHz
 - (b) 42.58 MHz
 - (c) 74.34 kHz
 - (d) 13.93 MHz
2. Calculate the magnetic field responsible for the polarization of 2.5×10^{-6} for ^{13}C at 298 K.
 3. 0.5 T
 4. 1.2 T
 5. 2.9 T
 6. 100 T

When the following equation is balanced, which of the following is true? $___ \text{MnO}_4^- + ___ \text{I}^- + ___ \text{H}^+ \rightleftharpoons ___ \text{Mn}_2^+ + ___ \text{IO}_3^- + ___ \text{H}_2\text{O}$

- (a) The $\text{I}^- : \text{IO}_3^-$ ratio is 3:1.
- (b) The $\text{MnO}_4^- : \text{I}^-$ ratio is 6:5.
- (c) The $\text{MnO}_4^- : \text{Mn}_2^+$ ratio is 3:1.
- (d) The $\text{H}^+ : \text{I}^-$ ratio is 2:1.

The ^{13}C spectrum of which isomer of C_6H_{14} has lines with three distinct chemical shifts?

- (a) hexane
- (b) 2-methylpentane
- (c) 3-methylpentane
- (d) 2,3-dimethylbutane

Of the following ionic substances, which has the greatest lattice enthalpy?

- (a) MgO
- (b) MgS
- (c) NaF
- (d) NaCl

True / False (10 marks)

Answer True or False

6. True or False: 0.050 M AlCl_3 has the highest ionic strength among the given solutions due to its complete dissociation into multiple ions.
7. True or False: Halogenated hydrocarbons are commonly considered safe and non-toxic in aquatic environments.
8. True or False: The polarization of a proton is 4.126×10^{-3} at 0.335 T and 2.142×10^{-6} at 10.5 T at 298 K.
8. True or False: The ^{55}Mn nuclide has five allowed energy levels for its nuclear states.
9. True or False: Argon has a lower boiling point than neon due to its stronger inter-molecular forces.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the procedures that minimize the influence of random errors on measured results in chemistry, focusing on signal averaging and averaging results from multiple samples.
12. Discuss the EPR spectrum of a rigid nitronyl nitroxide diradical with $J \ll a$, focusing on the factors that result in the appearance of nine lines. What does this indicate about the system?

End of Paper